

FIREFLY DRONE DESIGN

Cloud-Linked Autonomous Wildfire
Detection Architecture

University of California, Santa Cruz

Christian Li • Kenny Wang • Kevin Yao • Matin Sobhani • Nick Lafredo • Parsa Rezvani • Ryan Yu



EXECUTIVE SUMMARY

Bridging the Global Infrastructure Gap

Over **99%** of global forest canopies lack pre-set early detection hardware. Traditional methods rely on satellite passes or civilian reporting, often discovering fires only after they breach the canopy.

FireFly provides a continuous, closed-loop architecture coordinating automated edge hardware with scalable cloud resources to suppress spot-fires within minutes of ignition.

Core Objectives

- Zero-Infrastructure Deployment
- Dual-Spectral Heat Isolation
- Real-time Cloud Triangulation
- Scalable Regional Monitoring



CRITICAL NEED & VISION

The Problem: Remote high-risk areas lack localized sensors. Human vision is limited by smoke density and topographical blind spots.

The Solution: Deploy a rapidly mobilized autonomous network that transmits thermal and chemical telemetry to ground stations, alerting operators before thresholds are crossed.

*"ACT BEFORE THE FIRE BECOMES
UNCONTAINABLE."*



HEAVY-LIFT HARDWARE PLATFORM

950mm Industrial Configuration

Built for industrial-grade endurance and environmental survivability in disaster zones.

- **Frame:** 950mm diagonal carbon-fiber weave.
- **MTW:** 22-pound (10kg) Maximum Takeoff Weight.
- **Propellers:** Oversized 21-inch hollow-core blades.
- **Protection:** IP67 sealed fuselage with aerogel thermal shielding.



DUAL-SENSOR VISION

Long-Wave Infrared (LWIR): Pierces through opaque smoke plumes to locate the ignition seat. Operates in the $8\mu\text{m}$ to $14\mu\text{m}$ spectrum.

Optical Verification: High-definition RGB stream used by YOLOv8 for pattern recognition and smoke characterization.

Fusing thermal spikes with visual smoke textures reduces false positives by **85%**.



Goal

- Deploy a rapidly mobilized monitoring system for remote, high-fire-risk terrain
- Collect temperature, humidity, smoke, and visual data where sensors do not exist
- Transmit telemetry to a ground station and alert operators when fire is present
- Reduce detection delay so responders can act before a fire becomes uncontrollable



Personas

- Marcus, Wildfire Operations Coordinator
Needs fast deployment, reliable alerts, and minimal training for field crews.
- Elena, Ecology Researcher
Needs repeatable environmental data collection in remote areas.
- U.S. Forest Service Regional Office
Needs scalable monitoring for roadless forest regions with limited infrastructure.



Design Goals

- Triangulate fire location within acceptable GPS error.
- Support single-drone running-fix localization when only one drone is available.
- Merge repeated fire detections so the map does not flood with duplicate pins.
- Validate coordinate math and telemetry packets before updating system state.
- Keep frontend updates under the 500 ms polling window.



Principle Feature: Detection-to-Alert Pipeline

Sense -> Detect -> Localize -> Alert

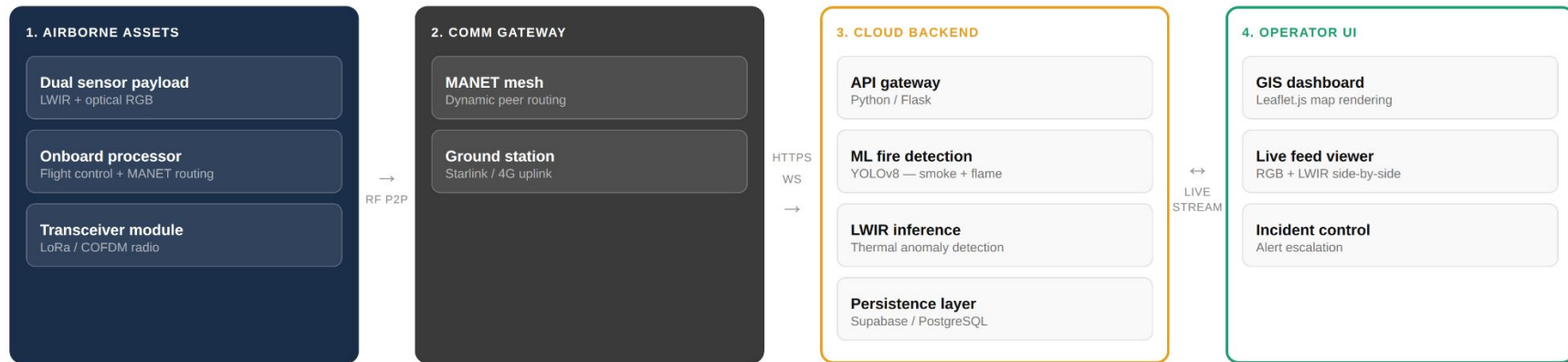
- ESP32-CAM streams visual input.
- YOLOv8/OpenCV detects fire and smoke offboard.
- Backend estimates fire coordinates using bearing and GPS data.
- Leaflet dashboard displays fire pins, drone markers, alerts, and live status.



System Schematic

HIGH-LEVEL DESIGN BLOCK DIAGRAM

End-to-end data pipeline and component relationships



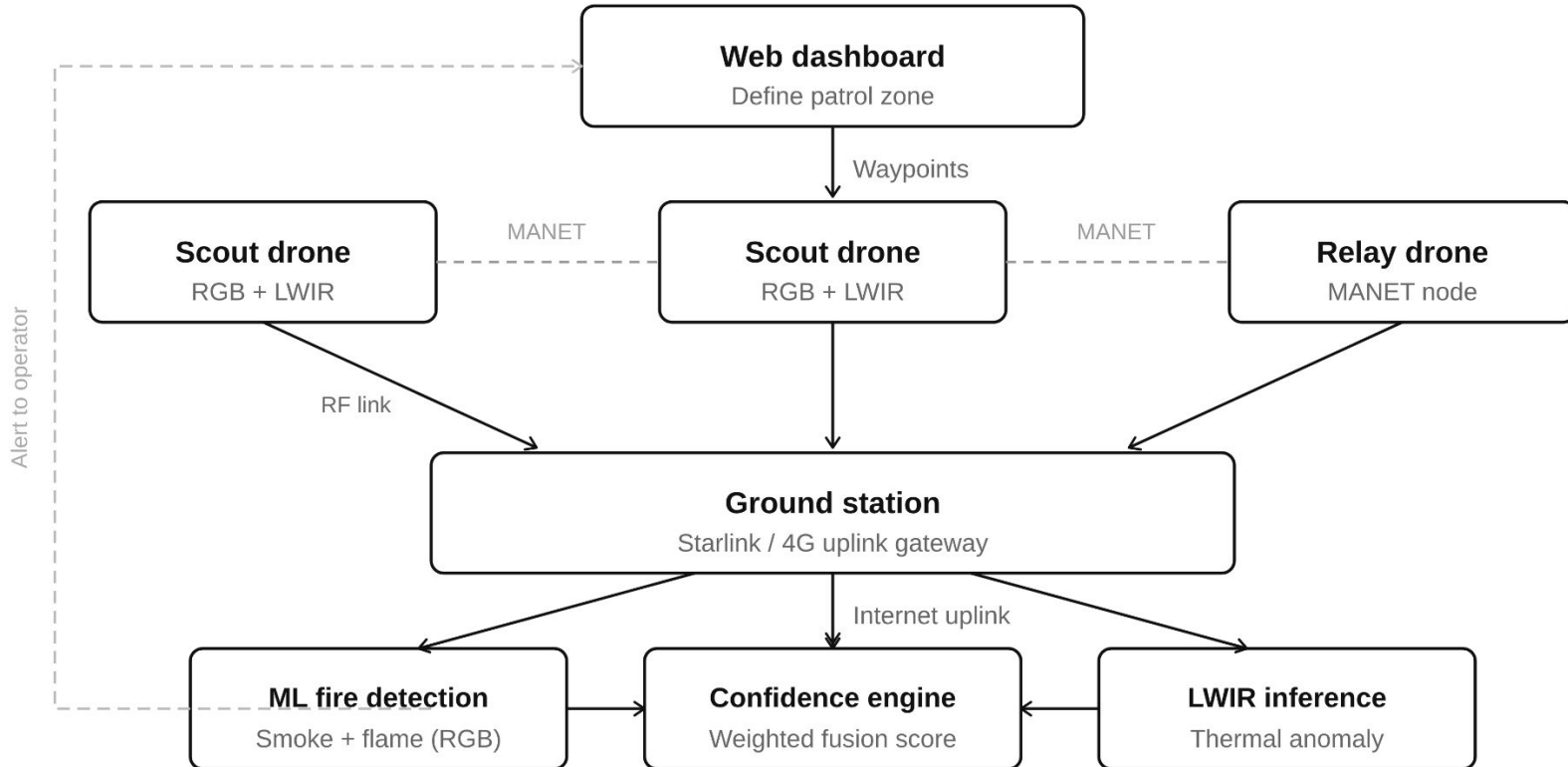
Data flows continuously left to right. Operator commands and waypoint updates transmit back down the pipeline.

● Hardware edge ● Network layer ● Cloud infrastructure ● Frontend UI

DETAILED DATA FLOW



System Schematic cont.



Prototype

- ESP32-C3 telemetry and identity node.
- ESP32-CAM visual node with MJPEG stream.
- Python YOLOv8/OpenCV vision worker.
- Flask backend with fire ingestion, telemetry state, and fire merging.
- Leaflet dashboard with map pins, bearing lines, alerts, and camera preview.

Results

- Fire TPR: 90.3%
- Smoke TPR: 85.7%
- Mean telemetry latency: 318 ms
- Indoor triangulation error: 2.1 m
- Simulated cruise endurance: 51 min



Thank you

